

Homework (Carolina Reaper)

- Q1.** Simplify $\sqrt{64x^3}$ in terms of x .
(A) $8x^{3/2}$
(B) $8x^{3/3}$
(C) $8x$
(D) $4x^{3/2}$
(E) $4x^{3/3}$
- Q2.** A painter mixes $\sqrt{128}$ liters of paint for an art project. How many liters of paint does the painter mix in simplest radical form?
(A) $8\sqrt{2}$
(B) $16\sqrt{2}$
(C) $4\sqrt{2}$
(D) 8
(E) 16
- Q3.** What is the simplified form of $\sqrt[3]{-64x^3}$?
(A) $-4x$
(B) $4x$
(C) $-4x^2$
(D) $-2x$
(E) $2x$
- Q4.** A music producer creates a rhythm with a frequency of $\sqrt{121}$ Hz. What is the frequency in simplest radical form?
(A) 11
(B) $11\sqrt{11}$
(C) $\sqrt{11}$
(D) $11^{1/2}$
(E) $11^{3/2}$
- Q5.** A canvas has dimensions $\sqrt{256}$ inches by $\sqrt{16}$ inches. What are the simplified dimensions?
(A) 16×4
(B) 4×4
(C) 8×2
(D) 16×2
(E) 8×4
- Q6.** What is $\sqrt[3]{8x^3}$ in simplest form?
(A) $2x$
(B) $2x^2$
(C) $2x^3$
(D) x
(E) x^2
- Q7.** A guitarist uses a string with a length of $\sqrt{900}$ cm. What is the length of the string in simplest radical form?
(A) 30
(B) $30\sqrt{2}$
(C) $10\sqrt{3}$
(D) 10
(E) $3\sqrt{10}$
- Q8.** Simplify $\sqrt[4]{81x^4}$.
(A) $3x$
(B) $9x^2$
(C) $3x^2$
(D) $9x$
(E) $3x^{1/2}$

Open Ended Questions (FRQ)

- Q9.** A dance troupe is performing a routine that involves $\sqrt{361x^2}$ steps in the x -direction. Simplify this expression and then convert it into rational exponent form. Show all work and explain reasoning.
- Q10.** Given $f(x) = \sqrt[3]{8x^3}$, convert this function to rational exponent form and simplify $f(x)$ to find the value of $f(4)$. Provide all steps and calculations.

Chef's Tips

- Tip 1: Encourage students to check for extraneous solutions when necessary.
- Tip 2: Make sure to review the conversion between radical and rational exponent forms before starting this homework.
- Tip 3: Provide additional challenges that involve real-world applications of radicals and rational exponents, especially in the context of Arts Music.